

Prediction of Diabetes (Pima Indian Females)

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Introduction

Indigenous Pima Indians of Arizona have the highest recorded rates of diabetes in the world¹. The Pima Indians Diabetes dataset² contains information about 768 Pima Indian women over 21 years of age and is used to predict onset of diabetes within 5 years. The dataset consists of eight predictor variables, and a binary outcome variable indicating if diabetes is diagnosed (yes = 1) or not (0). Physiologically impossible zero values are treated as missing and replaced using median imputation. In total 48.7% of insulin and 29.6% of skin thickness (ST) values were replaced. Other nonsensical values accounted for <4.6% of the respective variable. One record was removed due to impossible skin thickness, but other excessive values remained within possible ranges and were retained. Glucose and Blood Pressure exhibit symmetric distributions for both groups (fig. 1). BMI, Age Diabetes Pedigree Function (DPF) and pregnancies are right skewed. The diabetic group has a rightward shift compared to non-diabetics for age, glucose and BMI indicating that higher values are associated with increased diabetes risk. No obvious visual separation between outcomes for other variables. The distribution of insulin and ST are heavily influenced by imputed values.

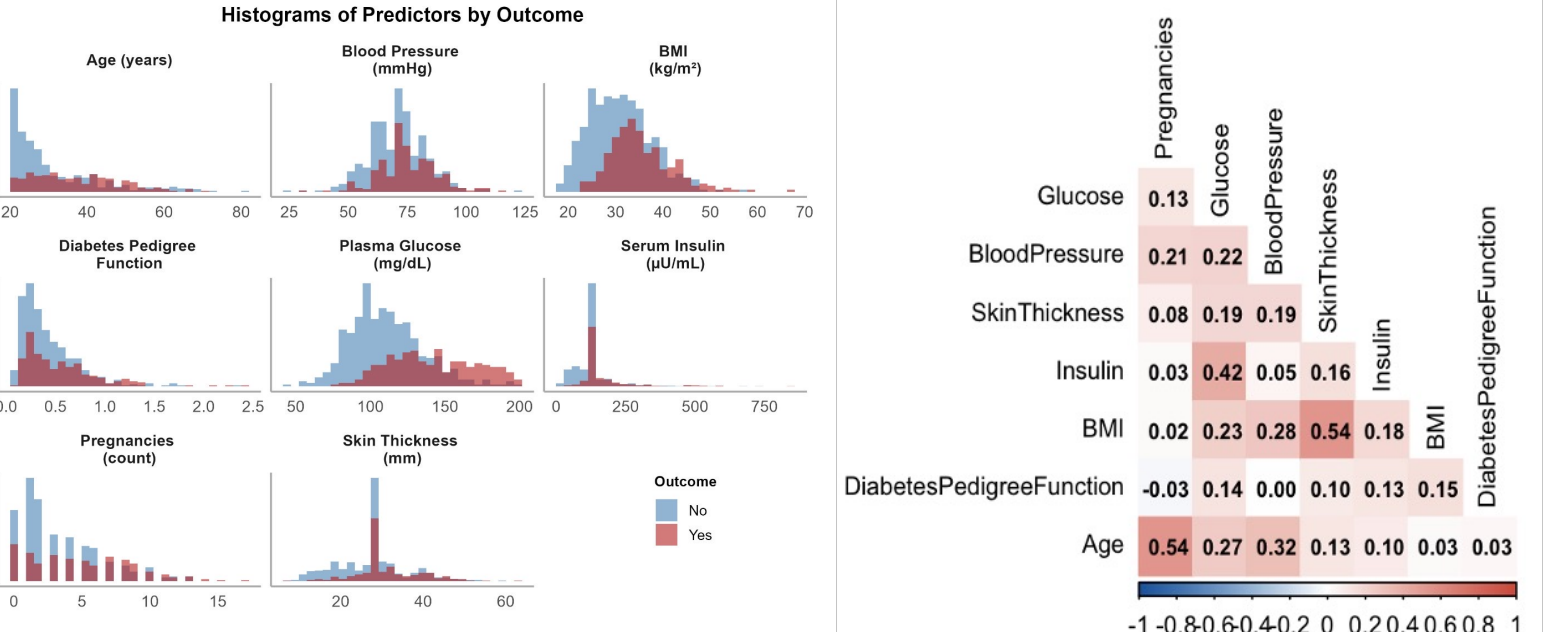


Figure 1. Distribution of the cleaned dataset 'classified' by whether a patient is diagnosed with diabetes (yes) or not.

Imputation/Exclusion: Correlation was low to moderate (fig. 2) with the largest correlations between BMI/ST (0.54), age/pregnancies (0.54) and glucose/insulin (0.42). Multicollinearity was insignificant for all variables. Due to the high percentage of imputation required for insulin and ST these variables were dropped from the binomial logistic regression and decision trees models.

Logistic Regression (Simple & Binomial)

Simple: Individual simple logistic regression models were fitted to predict outcome from each predictor variable. Metrics used to compare models were area under the curve (AUC) and accuracy. The best performing predictive variable using the validation set method was dependent on how the data was split into training and testing sets. The 10-fold cross validation (CV) method was stable with glucose consistently having the highest AUC and accuracy compared to the other variables. For a threshold of 0.5 (fig. 3), this model accurately predicts the outcome for 70.6% of patients. A sensitivity of only 0.45 indicates that over half the patients with diabetes are misclassified. To accurately diagnose 95% of true positive cases would result in an overall predicted diabetes rate of ≈82% which is significantly larger than the true positive rate (35%). In this context, a model that accurately predicts the presence of diabetes is desirable. However, this simple model does not sufficiently differentiate between patients with and without diabetes to be useful.

Binomial: Three methodologies (forward selection, backward elimination, stepwise regression) are all converged to the same set of six predictors: glucose, BMI, pregnancies, diabetes pedigree function (DPF), age, and blood pressure. Those variables can be identified regardless of the chosen strategy, which signal strong predictive values. Skin thickness and insulin were consistently excluded from all three models, the exclusion can be due to lack of predictive value, and high proportion of zero before imputation. The AIC score of 577.72 indicates good discriminative ability, with glucose and BMI functioning as the strongest predictors. Since stepwise regression offers greater flexibility by adding and removing predictors per step, lower sensitivity to the order of entry, and wider acceptance in biomedical domain, the stepwise model was retained for subsequent analysis.

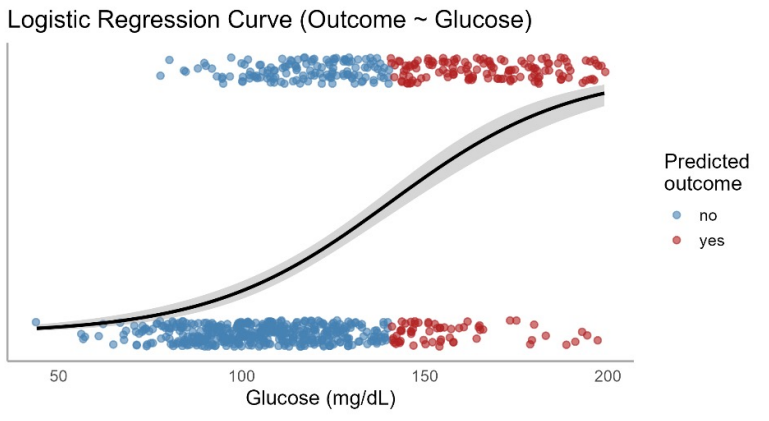


Figure 3. The logistic regression curve (black) and 95% confidence intervals (grey) for glucose as a predictor of diabetes using a 10-fold CV method. Predicted outcomes are based on a threshold of 0.5.

Variable	OR	95% CI	Interpretation
Glucose	1.039	(1.031,1.048)	↑Risk(3.9% per mg/dL)
BMI	1.094	(1.059,1.131)	↑Risk(9.4% per unit)
Pregnancies	1.123	(1.049,1.205)	↑Risk(12.3% per pregnancy)
DPF	2.017	(1.037,3.977)	↑Risk(doubles)
Age	1.019	(0.998,1.040)	Less significant
Blood Pressure	0.986	(0.968,1.005)	Less significant

Table 1. The Odd Ratio Table based on stepwise model to quantify the strength and direction of the association between a predictor variable and the diabetes outcome. All predictors except age and blood pressure had 95% confidence intervals without containing 1, confirming their statistical significance.

Decision Tree

As shown in the Figure 4,¹ original decision tree initially split on Glucose, showing that glucose was the most powerful variable of separating diabetes and non-diabetes cases. This would be relevant since glucose is directly linked to the risk of having diabetes. The later splits were BMI, Age, Diabetes Pedigree Function, Blood Pressure, and Pregnancies which are also practical health indicators to be measured or collected before doing more complex testing.

The original tree achieved 72.6% accuracy, 62.3% sensitivity, 78% specificity, and AUC = 0.755. This indicates that the model was fairly good in general and was more precise to recognize non-diabetic than diabetic patients. However, it was still missing 20 true diabetes cases. This is significant in a healthcare screening context since a false negative situation might cause a delay in follow-up testing or treatment. So, the original tree can be useful to describe the patterns of diabetes risks, but it cannot be considered as a final diagnosis model.

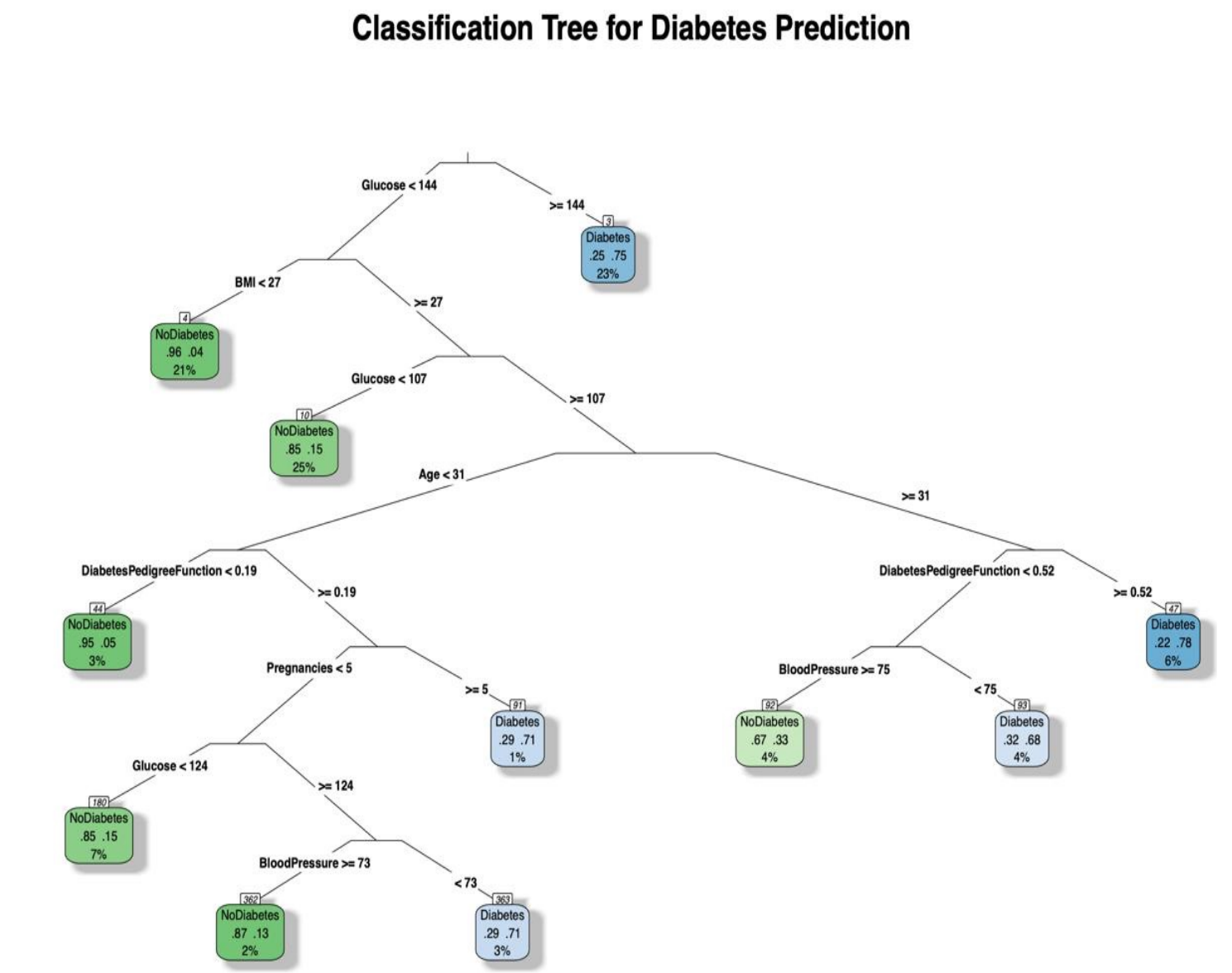


Figure 4. Original decision tree model for diabetes screening

From the Figure 5, the pruned tree was a simplified version of the original tree as it cut some off some of the smaller branches. It retained the primary structure, with Glucose <144 as the first split, then BMI, Glucose<107, and Age. This simplifies the model and makes it more appropriate to explain the key risk patterns.

The pruned tree got an accuracy of 73.2%, a sensitivity of 67.9%, a specificity of 76%, and AUC = 0.757. Compared to the original tree, the pruned tree was slightly more accurate, more sensitive, and fewer false negatives (17 instead of 20 of the original tree). This showed that pruning did not make the model weak. Instead, it simplified the tree and enhanced the capacity to discriminate cases of diabetes. However, its specificity was slightly lower, meaning it produced more false positives, where some non-diabetes patients were incorrectly flagged as diabetic.

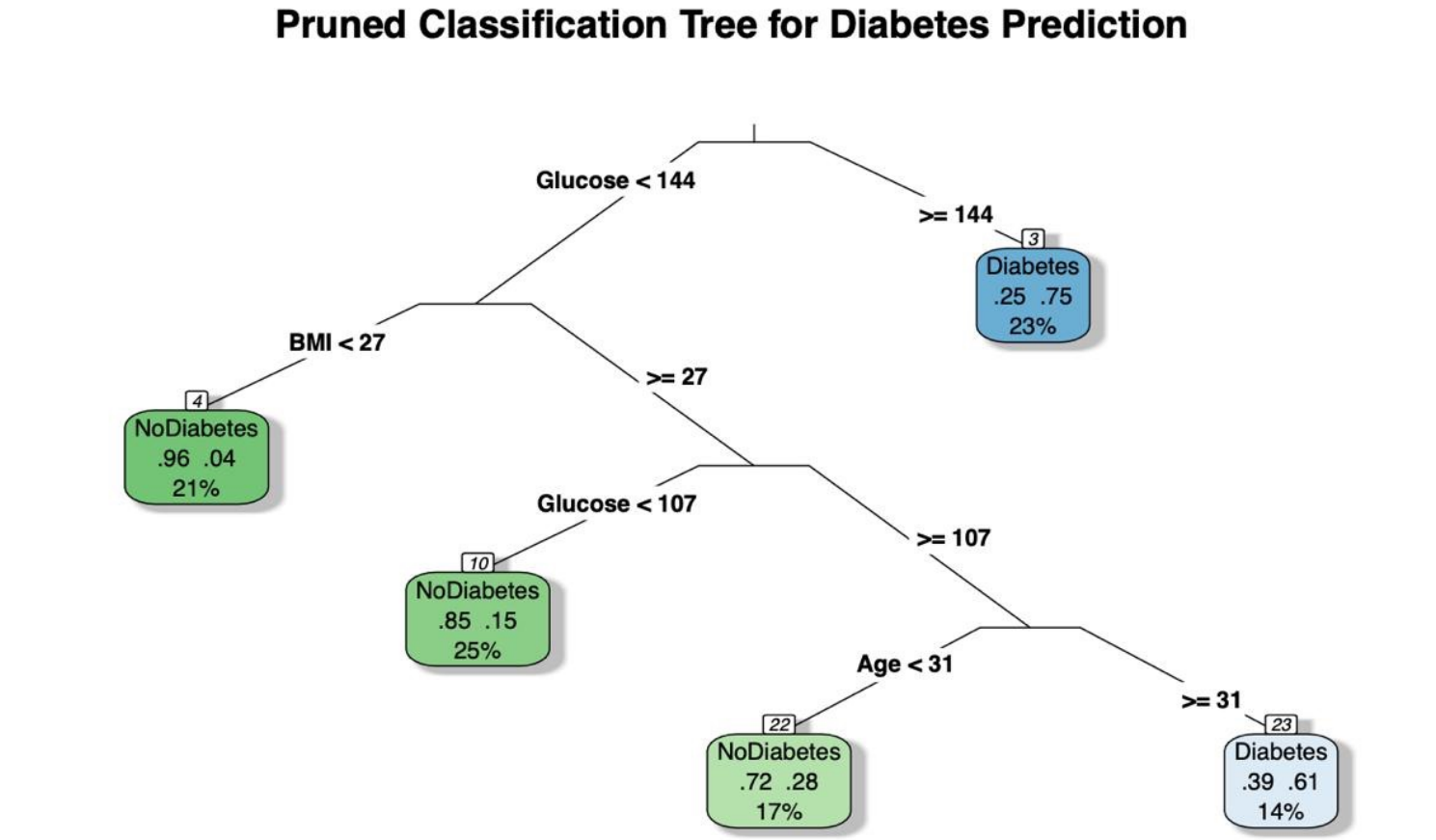


Figure 5. Pruned decision tree model for diabetes screening

Thresholds Trade-off

Model	Metric	Threshold 0.30	Threshold 0.5	Change
Simple Logistic Regression	Accuracy	66.7%	70.6%	Increased
	Sensitivity	75.5%	45.3%	Decreased
	Specificity	62.0%	84.0%	Increased
	False Positives	38	16	Decreased
	False Negatives	13	29	Increased
	Accuracy	73.2%	72.5%	Decreased
Binomial Logistic Regression	Sensitivity	83.0%	47.2%	Decreased
	Specificity	68.0%	86.0%	Increased
	False Positives	32	14	Decreased
	False Negatives	9	28	Increased

Table 2. Effects of increasing the threshold from 0.30 to 0.50

The logistic regression models were tightened by increasing the threshold from 0.30 to 0.50. It lowered false positives but raised the false negatives. In the binomial logistic regression model, the number of false negatives rose from 9 to 28, resulted in more true diabetes cases being missed. Therefore, the preferred value is 0.30 for diabetes screening, in which the consequence of missing true diabetes cases is more harmful than the extra testing of patients.

Model Comparison & Evaluation

Model	AIC	AUC	Accuracy	Sensitivity	Specificity	False Positives	False Negatives
Logistic Regression							
Simple	629	0.749	66.7%	75.5%	62.0%	38	13
Binomial	578	0.810	73.2%	83.0%	68.0%	32	9
Decision Trees							
Original		0.755	72.6%	62.3%	78.0%	22	20
Pruned		0.757	73.2%	67.9%	76.0%	24	17

Table 3. Model evaluation summary

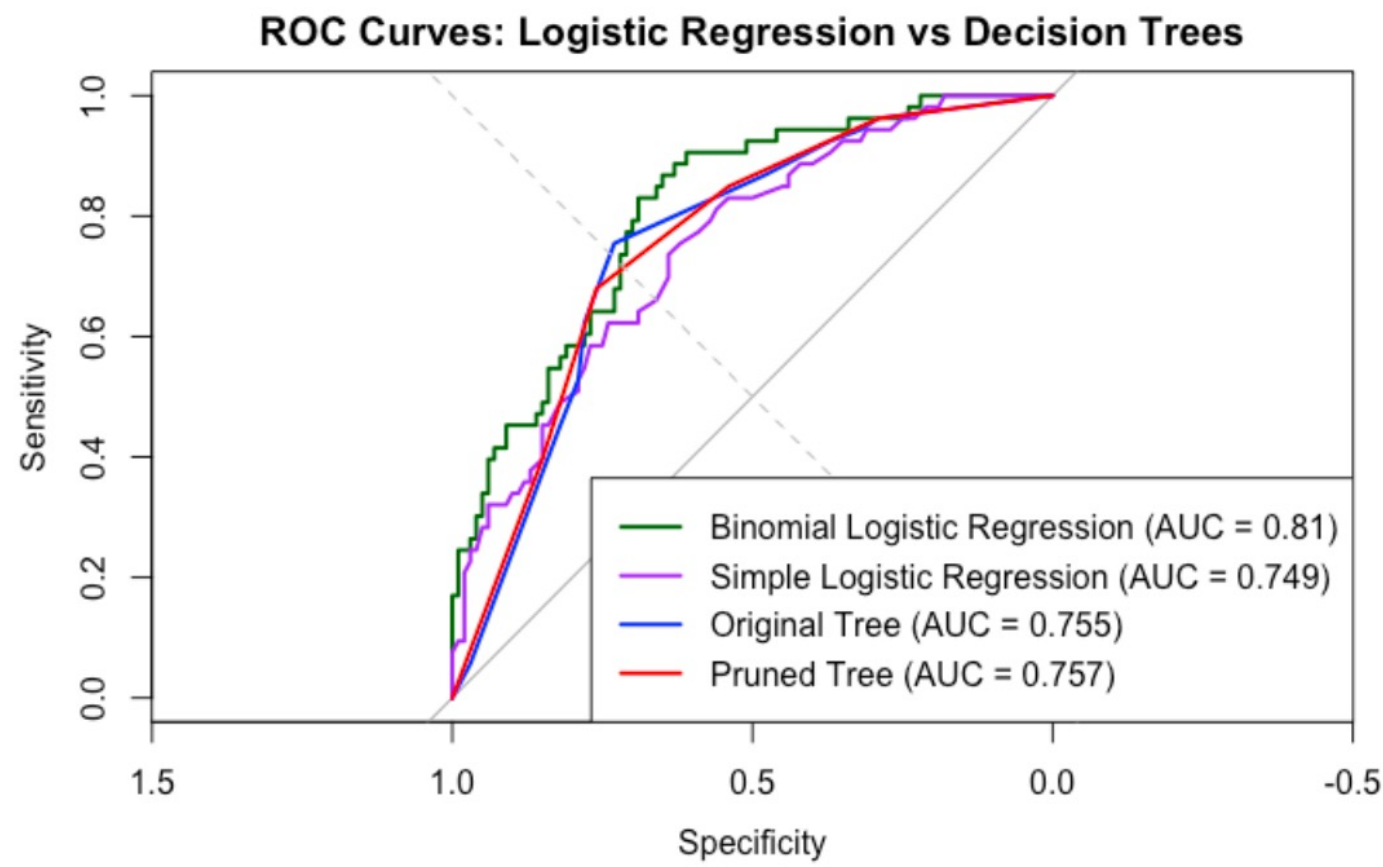


Figure 6. ROC curve comparison across models

The best screening model was the binominal logistic regression with the largest AUC = 0.810, the highest sensitivity = 83%, and the fewest number of false negative cases = 9. This is important since false negatives can result in missing true diabetes cases.

Result of simple logistic regression shows that glucose alone is useful, with AUC = 0.749 and sensitivity = 75.5%. However, it was the least accurate (66.7%), also has the highest false positives (38 positive cases that are incorrect), so glucose alone is not sufficient for the final model.

The decision trees were less predictive but easier to explain than the binominal logistic regression. Compared with the original tree (20 false negatives), the pruned tree reduced the number of false negatives to 17 and had a similar AUC (0.757). Overall, logistic regression will perform well for screening predictions, whereas the pruned tree is more useful for understanding diabetes risk patterns.

Non-invasive Screening Recommendations

Measuring plasma glucose concentration and insulin levels requires invasive blood testing, which is costly, time consuming and requires specialist equipment. We wanted to explore if simple, non-invasive measurements can be used to identify patients most at-risk of developing diabetes or having high glucose levels, who could then be referred for testing.

Multiple logistic regression to predict diabetes

A forward regression model identified BMI, age, DPF and pregnancies as most significant for predicting diabetes. The optimal ROC threshold of 0.265 resulted in a sensitivity of 0.84. Approximately 60% of patients were identified as being at risk of diabetes, and less than 6% of patients that did get diabetes were not correctly identified. Although this model hugely overestimates the true number of positive cases, it does a reasonable job of identifying individuals that should undergo further testing.

Multiple linear regression to predict glucose

Age, BMI and DPF are found to be the only significant variables in predicting glucose. However, they only explain ≈14% (R²) of the variability in glucose. Therefore, glucose levels are more heavily influenced by unknown factors, and this model cannot be used to reliably identify individuals who may have elevated glucose levels. Non-invasive variables are better at predicting the onset of diabetes than predicting glucose.

Conclusion

The strongest predictor was glucose, but the best performance can be achieved in the binominal logistic regression model, which included glucose together with BMI, pregnancies, diabetes pedigree function, age, and blood pressure. This showed that a broader health-risk profile is better at predicting diabetes risk, rather than glucose alone.

The 0.30 threshold was more suitable for screening purposes as this threshold had a lower rate of missed diabetes cases and a higher rate of false positive cases. This is acceptable in the case of healthcare screening because additional testing is safer than not catching actual diabetes cases.

The decision trees were less predictive, but simpler to explain, particularly the pruned tree. Overall, the models can serve as screening tools for early detection, but not the final diagnostic tools.

Limitations

- Several limitations existed in this study that should be acknowledged:
- Unique ID:** There is no identification attribute, so it is impossible to know if the dataset contains only unique entries or if some patients are represented multiple times.
- Insulin/Skin Thickness Data:** The large imputations of insulin (48.7%) and skin thickness (29.6%) can possibly misrepresent the true distributions and underestimate the variances. This might limit to explore the significance of variables.
- Age:** The age restriction excludes younger participants (< 21 years old), limiting the model's applicability to early-life risk evaluation.
- Gender:** The absence of male participants, means the model's predictive performance for men remains unknown.
- Ethnicity:** The data was collected from the Arizona group of Pima Indians. The Journal of Clinical Investigation has stated "less variation in this region was found in Pima Indians than in other racial groups previously studied"³. This finding demonstrates that a substantial proportion of genetic variation in this population is distinct and therefore not represented in studies of other ethnicities.
- External Validation:** Without external validation on more independent, diverse populations, the remains unknown.
- Timeframe:** The dataset reflects the medical conditions around 1988 rather than nowadays. Therefore, future studies should validate the model using more diverse datasets across broader and more representative demographic groups. Additionally, data was captured during a medical examination every two years from 1965–1988, however, there is no indication when each measurement was made.

References

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